

Python Fundamentals

What is Python?

Python is a **high-level, interpreted, general-purpose programming language** used to build software applications, automate tasks, analyse data, create websites, develop AI solutions, and much more.

Python follows the principle:

"Humans First, Computers Second"

It was designed to be:

- Simple
- Readable
- Easy to learn
- Productive

Why Python is Popular?

Python has become one of the most widely used programming languages because:

1. Simple English-like Syntax

Python code looks similar to normal English sentences.

Example:

```
print("Hello World")
```

2. Less Code

Python allows developers to perform more work with fewer lines of code.

3. Large Ecosystem

Thousands of libraries are available for almost every task.

4. Cross Industry Adoption

Python is used in:

- ✓ Web Development
- ✓ Artificial Intelligence

- ✓ Machine Learning
- ✓ Deep Learning
- ✓ Automation
- ✓ Data Science
- ✓ Data Analytics
- ✓ Business Applications
- ✓ Game Development
- ✓ IoT
- ✓ Testing Automation
- ✓ Education
- ✓ Software Development
- ✓ Visualization
- ✓ Audio & Video Processing

Applications of Python

Python is used in many industries.

Web Development

Frameworks:

- ✓ Django
- ✓ Flask
- ✓ FastAPI
- ✓ Artificial Intelligence

Examples:

- ✓ Chatbots
- ✓ Recommendation Systems
- ✓ Computer Vision

Libraries:

- ✓ TensorFlow
- ✓ PyTorch
- ✓ Scikit-learn

Data Science

Libraries:

- ✓ Pandas
- ✓ NumPy
- ✓ Matplotlib
- ✓ Seaborn

Automation

Python helps automate repetitive tasks.

Examples:

- ✓ File handling
- ✓ Report generation
- ✓ Email automation
- ✓ Testing

Testing

Automation tools:

- ✓ Selenium
- ✓ PyTest

IoT Applications

Python is widely used with:

- ✓ Raspberry Pi
- ✓ Embedded Systems

Game Development

Library:

✓ Pygame

Example:

✓ 2D Games

History of Python

Inventor, “**Guido van Rossum**”

Developed at CWI, (Centrum Wiskunde & Informatica) Netherlands

Timeline, 1989 - Project started

1991 - Python 0.9.0 released

1994 - Python 1.0 released

2000 - Python Software Foundation established, Open-source community growth

Origin of the Name Python:

Python was named after the British comedy show: “*Monty Python's Flying Circus*”

It was not named after the snake.

Computer Languages

What is a Computer Language?

Computers do not understand human languages.

Computers understand only:

Binary: 0 and 1

This is called: Machine Language

Binary System

0 = OFF

1 = ON

Similar to an electrical switch.

Types of Programming Languages

1.Low Level Languages

Examples:

Machine Language

Assembly Language

Characteristics:

- ✓ Hardware dependent
- ✓ Faster execution
- ✓ Difficult for humans

2.High Level Languages

Examples:

- ✓ Python
- ✓ Java
- ✓ Ruby
- ✓ C++

Characteristics:

- ✓ Human friendly
- ✓ Easy to write
- ✓ Easy to maintain
- ✓ Portable

Compiler vs Interpreter

Compiler

Converts the entire program into machine code at once.

Examples:

- ✓ C
- ✓ C++

Interpreter

Converts code line by line.

Example:

- ✓ Python

Advantages:

- ✓ Easier debugging
- ✓ Immediate execution

Why Python is Easy?

Simple Syntax

Python uses readable code.

No Heavy Symbols

No semicolons are mandatory.

Less Coding

More productivity.

Readability

Code resembles normal sentences.

Environment Setup

There are two ways to learn Python.

Option 1 – Install Python Locally

Required Tools, Download from: [Python.org](https://python.org)

IDE:

- ✓ PyCharm
- ✓ VS Code
- ✓ Spyder
- ✓ IDLE

Benefits

- ✓ Syntax highlighting
- ✓ Error detection
- ✓ Auto completion
- ✓ Debugging support

Online Platforms

Useful for beginners.

Examples

- ✓ OneCompiler
- ✓ Google Colab

Advantages

- ✓ No installation
- ✓ Accessible anywhere
- ✓ Free practice environment